Suman Cha

Integrated M.S./Ph.D. Student in Statistics and Data Science ◇ Yonsei University

oldrain123@yonsei.ac.kr

Website GitHub Google Scholar

EDUCATION

Yonsei University *March 2021 – February 2027 (expected)*
Integrated M.S./Ph.D. in Statistics and Data Science
Co-advised by Prof. Hyunjoong Kim and Prof. Ilmun Kim.

Yonsei University *March 2015 – February 2021*
B.S. in Applied Statistics

RESEARCH AND INDUSTRY EXPERIENCE

KRAFTON *June 2025 – March 2026*
AI Applied Research Intern, User Modeling AI Team
- PUBG MMR Renewal: MMR Algorithm Development
- PUBG E-sports Real-time Win Probability Prediction: Model Development

Yonsei University *January 2025 – December 2026*
Researcher, Wastewater Treatment Process Intelligence
Developed time-series forecasting models for wastewater treatment processes.

Yonsei University *September 2022 – February 2025*
Researcher, Basic Research Lab (BRLab)
Conducted research on conditional two-sample testing and participated in biweekly seminars.

PUBLICATIONS AND MANUSCRIPTS

[1] **Suman Cha**, Seongchan Lee, Dohyun Ko, and Hyunjoong Kim†, "TabFUSE: Feature-Wise Unified Specialization with Cross-Column Exchange for Mixed-Type Tabular Flow Matching." *Submitted to AISTATS 2027.*
[arXiv · Code]

[2] **Suman Cha**, Ahyoung Lee, and Hyunjoong Kim, "Semi-Supervised Tabular Learning with Class-Specific Reconstruction and Reliable Pseudo-Label Selection." *Submitted to AISTATS 2027.*
[OpenReview]

[3] Jungmin Kim*, **Suman Cha***, Seunghan Lee*, Seunghyun Kim, Junekyu Park, Jiyoung Lim, Anna Lee, and Hyuck Lee†, "Real-Time Win Probability Prediction in Battle Royale Games via Survival Analysis." *Submitted to KDD 2027 Applied Data Science Track.*

[4] **Suman Cha***, Seongchan Lee*, Antonin Schrab, and Ilmun Kim, "More Permutations Do Not Always Increase Power: Non-monotonicity in Monte Carlo Permutation Tests." *Accepted at Statistical Science.*
[arXiv · Code]

[5] **Suman Cha** and Hyunjoong Kim, "Learning Majority-to-Minority Transformations with MMD and Triplet Loss for Imbalanced Classification." *Information Sciences 123929 (2026).*
[Publisher · arXiv · Code]

[6] Seongchan Lee*, **Suman Cha***, and Ilmun Kim, "General Frameworks for Conditional Two-Sample Testing." *Accepted at Biometrika.*
[arXiv · Code]

* Equal contribution. † Corresponding author.

OPEN RESEARCH SOFTWARE

Generative and Imbalanced Learning

- **TabFUSE**. Reproducibility pipeline with SHA-256-verified data reconstruction and command-line workflows for training, reproduction, dependence studies, replay, and evaluation. (Publication 1.)
- **MOMS**. Parametric majority-to-minority transformation using MMD for global alignment and triplet loss for local boundary awareness, evaluated on 29 datasets. (Publication 5.)

Statistical Testing

- **Cond2ST**. R package and reproducibility code for two general conditional two-sample testing approaches based on conditional independence tests and density-ratio testing. (Publication 6.)
- **mc-perm-power**. R, Rcpp, and C++ reproducibility code for the non-monotone unconditional power and integer-threshold sawtooth behavior of standard non-randomized Monte Carlo permutation tests. (Publication 4.)

TEACHING EXPERIENCE

Teaching Assistant, *Multivariate Analysis*
Yonsei University — *March 2025 – August 2025*

Teaching Assistant, *Mathematical Statistics*
Yonsei University — *March 2024 – August 2025*

Teaching Assistant, *Theory and Applications in Machine Learning*
Yonsei University — *September 2021 – December 2021*

SELECTED TALKS AND AWARDS

Talks

2025 **Oral presentation**, JSM 2025.
General Frameworks for Conditional Two-Sample Testing

2024 **Oral presentation**, KDMS 2024.
Contrastive Learning for Two-sample Tests

Awards

2024 **Best Paper Award**, KDMS 2024.
2022 **Excellence Award**, Yonsei Data Science Competition - KCB Project.
2021 **Excellence Award**, Yonsei Data Science Competition - Hyodol Project.
2019 **Encouragement Award**, Sony Pictures - Movie Audience Prediction.
2019 **Excellence Award**, Yonsei Data Science Competition - PACKUS.
2019 **Excellence Award**, Yonsei Data Science Competition - Embrain.

TECHNICAL SKILLS

Programming and ML — Python, PyTorch, R, C++
Data platforms — Databricks, PySpark